

SEIZURE

is a transient occurrence of signs and/or symptoms resulting from abnormal excessive or synchronous neuronal activity in the brain.

Simple seizure: -no alteration in consciousness.

oComplex seizure: -no alteration in consciousness.

EPILEPSY

- is present when two or more unprovoked seizures occur in a time frame of longer than 24 hours.

CAUSES OF SEIZURE

- 1. Idiopathic
- 2. CNS infections
 - Meningitis
 - Encephalitis
 - Brain abscess
 - Syphilis(tertiary)

- Metabolic
 - Electrolyte imbalance (calcium, sodium, magnesium, phosphorus)
 - Hypoglycemia
- 4. Structural
 - Head trauma
 - Brain tumor
 - Stroke
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- 5. Others

Sepsis

Drug abuse

Ingestion of toxins (accidental and non-accidental)

CLASSIFICATIONS OF SEIZURE

- Seizure can be classified as
 - o Focal: initial activation of a system of nervous limited to one part of hemisphere.
 - o Generalized: first clinical and EEG changes indicate synchronous involvement of all both hemisphere

TYPES OF FOCAL AND GENERALIZED SEIZURES:

- Tonic: increased tone or rigidity.
- • Atonic: flaccidity or lack of movement during convulsion.
- • Clonic: rhythmic fast muscle contraction and slightly longer relaxation.
- • Myoclonic: shock like contraction of a muscle < 50msec often repetitive.
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Febrile seizure

- a seizure that occurs at 38°C or higher, that are not the result of CNS infection or any metabolic imbalance, and in the absence of prior afebrile seizures.

Clinical features

- Elevated temperature ($>38^{\circ}\text{C}$)
- Convulsions usually brief and self limiting (usually <5 minutes, and may be prolonged for more than 15 minutes in complex febrile seizure) but may recur if temperature remains high
- o No neurological abnormality in the period between convulsions
- o Generally benign and with good prognosis

Treatment

- Give an antipyretic: paracetamol 15 mg/kg every 6 hours until fever subsides
- If convulsing: Give diazepam 500 micrograms/kg rectally (using suppositories/rectal tube or diluted parenteral solution) Maximum dose is 10 mg, Repeat prn after 10 minutes

Status epilepticus

- Status epilepticus generally refers to the occurrence of a single unremitting seizure with duration longer than 30 minutes or frequent clinical seizures without an interictal return to the baseline clinical state.

treatment

- Stabilization of the airway, maintenance of adequate ventilation (with oxygen administered as necessary), and circulatory support.

- Dextrose 10% 5 mL/kg children
- Diazepam as above in febrile convulsion,
repeated after 5-10 min If not responsive,
Phenytoin 15-18 mg/kg over 1 hour

- If no response consider Phenobarbital 10-15 mg/kg slowly IV then 10mg per kg bid

- A longer-acting antiepileptic drug should also be administered because of the relatively short duration of action of diazepam, Phenytoin is preferred over phenobarbital

